

3D PRINTING

Printing the Future

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ENGL 3153

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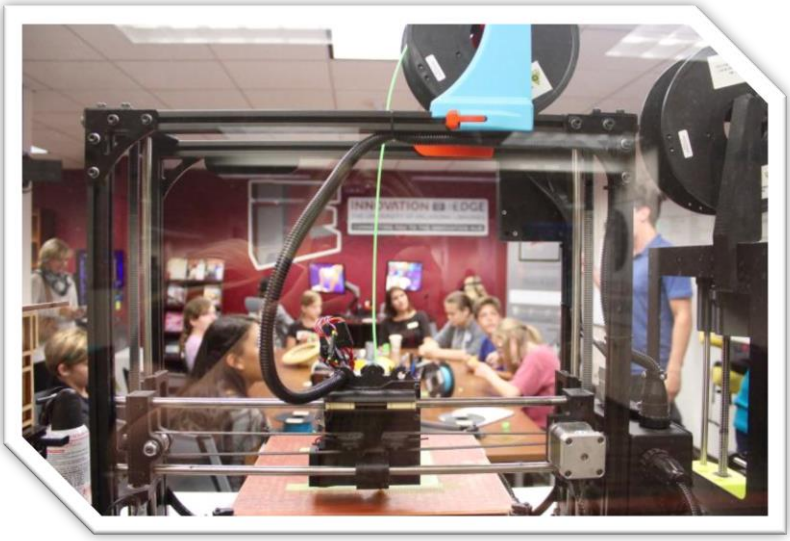
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Blog Post: 3D Printers at OU!

3D printers have existed for just about four decades now. In 1986, Chuck Hull patented an invention which allowed him to use UV light to harden plastic, by doing so in layers he was able to create 3D objects. Today's printers do not use UV light to print objects and devices. Instead, by heating plastics or other materials up, layers are extruded, and thus 3D creations are made!

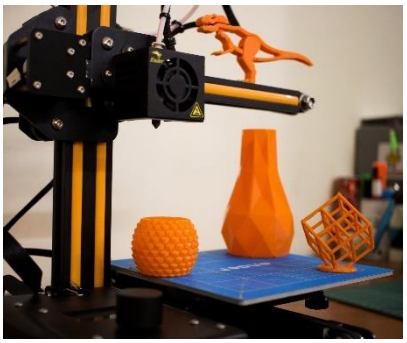
When Hull first came up with his creation, he predicted that it would take around 30 years for the printers to develop enough for them to find their way into homes (Hickey). And he was correct! Like the computer, 3D printers have increased in precision and power, while they decreased in size and cost. While 3D printers might not yet be in every house, many local libraries have 3D printers for the community to use. Learning more about 3D printing and modeling can be a fun hobby and teaches you valuable skills! MakerBot, a well renowned 3D printer company, supports teaching students how to 3D model and print their creations. Doing so helps teach critical thinking, collaborating, and helps to boost confidence ("7 Life Skills You Can Teach with a 3D Printer"). These skills aren't only valuable for younger students, but also can help adults visualize how to approach complex problems!

Here at the University of Oklahoma, we have a MakerSpace in our library. This space helps support students who want to explore 3D printing and print something of their very own! Not only can students print cool toys or models here, but students also often use this space to help with class or personal projects. We live in a three-dimensional world and 3D modeling can help us better understand and represent that space, expanding our horizons in an ever-growing search for knowledge. Stop by the OU's very own MakerSpace to learn more about 3D printing today!

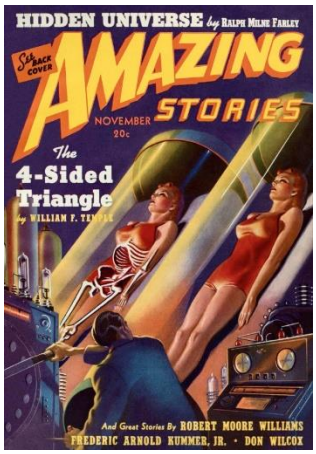


Cornerstone Content Post

We live in a three-dimensional world. We look at objects and our brains process what we and transforms it into an image. However, when we look at the world through a 2D lens, we lose some information in the transformation. Instead, many topics (engineering, medicine, entertainment, etc.) can benefit by storing information in a 3D format. By creating and printing 3D models, confusing problems can become straightforward. Sometimes an issue just needs to be approached from a different angle and what better way to see all of the angles than with three dimensions.



3D printing is the process of making a physical object from a three-dimensional digital design. This process is usually done by 'printing' thin layers of material over each other and fusing them. When most people think about 3D printing, they probably think about making little toys and figures to mess around with or display but its use has much more potential than that. 3D printing can be and is used in multiple different industries and applications. From healthcare, fashion, design, and even personal use, the versatility of 3D printing is immense.



One of the earliest mentions of creating objects via machines like 3D printers was 83 years ago in a novelette titled "The 4-Sided Triangle" by William F. Temple ("What Is the First Reference to a 3D Printer in Science Fiction?"). However, as fiction has turned to fact, it is important to make note of what is actually possible and what still remains outside of our current grasp. In 1986, Chuck Hull patented an invention which allowed him to use UV light to harden plastic, by doing so in layers he was able to create 3D objects. When Hull first came up with his creation, he predicted that it would take around 30 years for the printers to develop enough for them to find their way into homes (Hickey). And he was correct! Today 3D printers are more widely available and are used in both scientific development and for personal enjoyment.

As mentioned before, 3D printing is a type of manufacturing that works by laying down consecutive layers to create an object. This is called additive manufacturing. Additive manufacturing is the opposite of subtractive manufacturing where material is removed from a starting piece instead. Before you can print a 3D object, you must first obtain a 3D model. These models can be downloaded from websites like thingiverse.com and printables.com or they can be created. This model then gets loaded into computer software that "slices" it to the specific 3D printer's specifications. Essentially, as a 3D printer works in layers, the model must be "sliced" so that the printer knows what portions of the model go in the same layer. This helps to avoid print errors and maximize print efficiency ("What Is 3D Printing?").

3D printing provides many benefits. One main benefit of 3D printing is that it makes it possible to rapidly prototype of products. By being able to 3D model a part or object and then can

immediately test it in a cost-effective manner by printing it, designers gain a valuable perspective (McFadden). Not only does this technology provide many benefits to prototyping, but it also expands our technological boundaries.



One direction that 3D printing is expanding our horizons is up; as in, up towards space and the stars. While manned missions that orbit Earth are at a close distance and are able to be resupplied with parts, future missions that travel farther will need a different way to get replacement parts. As we journey back to the Moon and onwards to Mars, astronauts will need to be able to make their own replacement parts in space. Not only are routine replacements needed, but emergency situations may arise where astronauts need to make a new part quickly (Gaskill). 3D printing could just be the answer to the challenges that need to be overcome to make space travel possible. A project run through NASA has been experimenting with 3D printing in space on the International Space Station. “Analysis revealed that microgravity had no engineering-significant effects on the process, demonstrating that a 3D printer works normally in space and paving the way to new logistics systems for long duration missions. One of the first parts printed on the space station during 3D Printing in Zero G, a wrench, demonstrated it was possible to remotely send a design from the ground to a manufacturing system more than 200 miles above” (Gaskill).

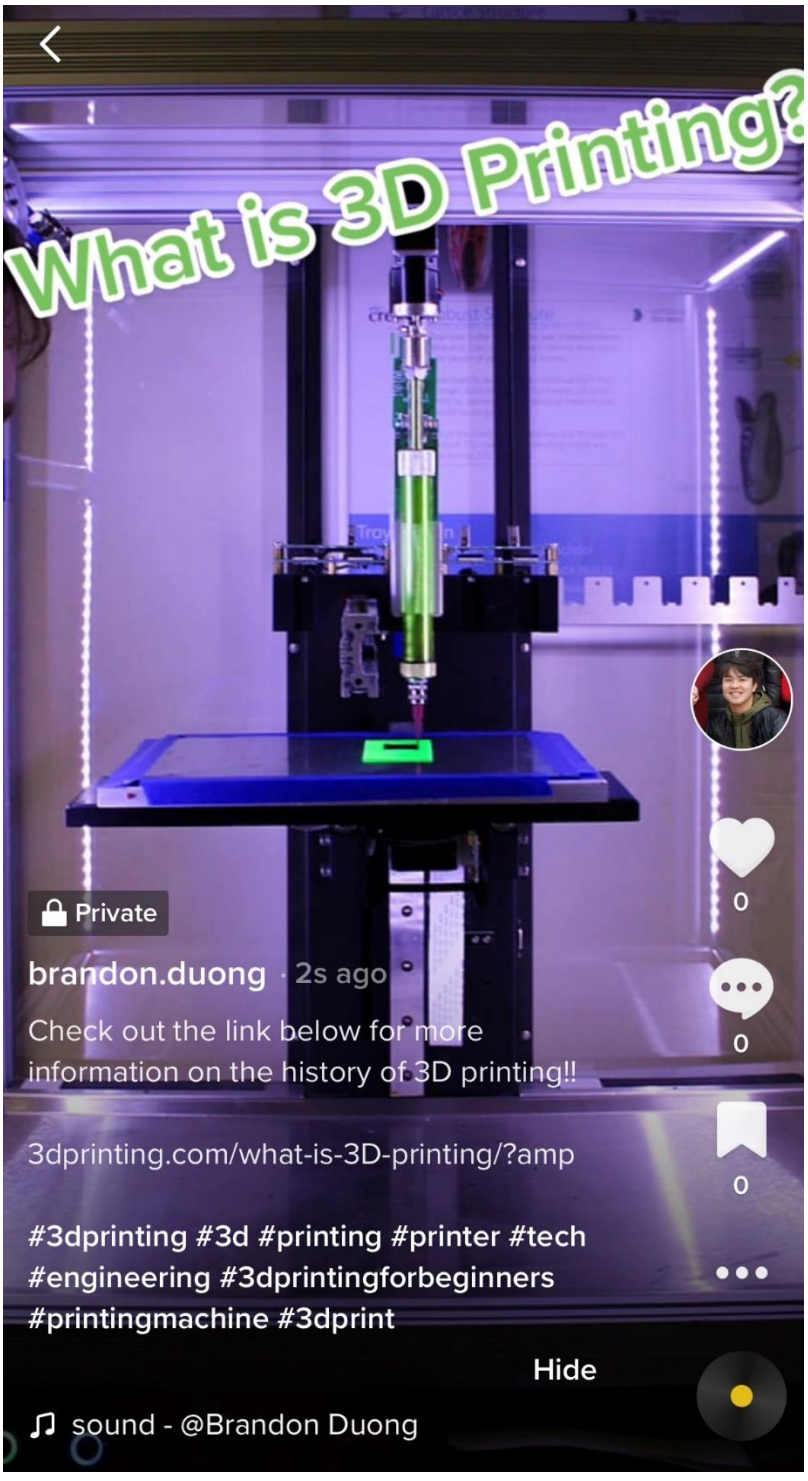


While it is great to think about the future and all that it holds, both here on Earth and beyond, you might be wondering, “Yeah, okay. But what can *I* do with 3D printing?” Great question! Let’s take a brief look into how you can learn more about 3D modeling and printing. As mentioned before, 3D models can be downloaded from websites, or can be created with open source software like www.blender.org. Blender is a great resource to get into 3D rendering in general. However, it can be used not only for 3D graphics but also to create 3D models and which can then be printed (Gregurić). But where to print your newly acquired/created 3D model? At the local library! Many libraries have already added or are working to add 3D printers to their facilities. With the use of a public library card, you might already have free access to a 3D printer (O'Neal).

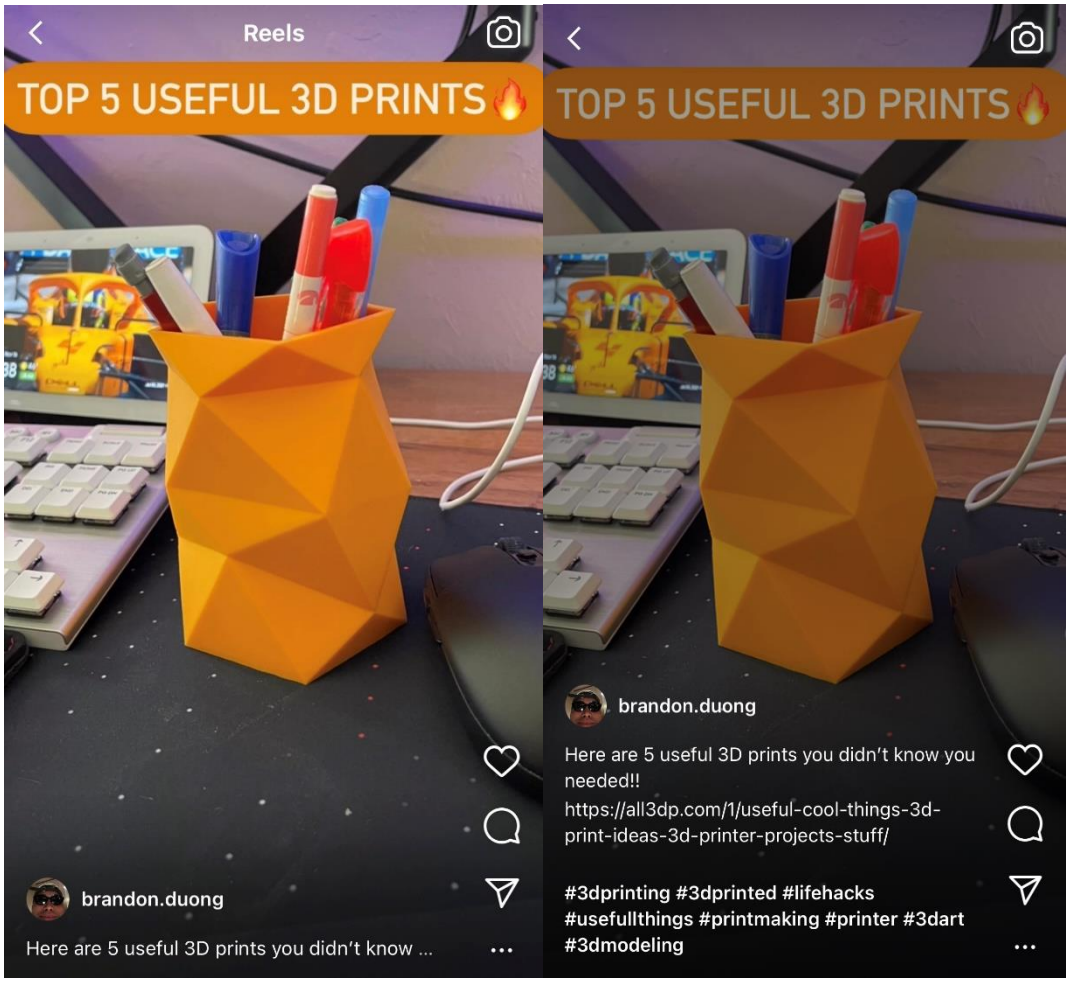
3D modeling and printing not only presents scientific breakthroughs and important opportunities, but it is also a fun hobby for anybody to pick up! A toy rocket that you print at a library or at home might not end up among the stars, but 3D printing most likely will! Scientific boundaries are there to be pushed, and 3D printing provides a path to push them on all fronts.

Social Media Posts

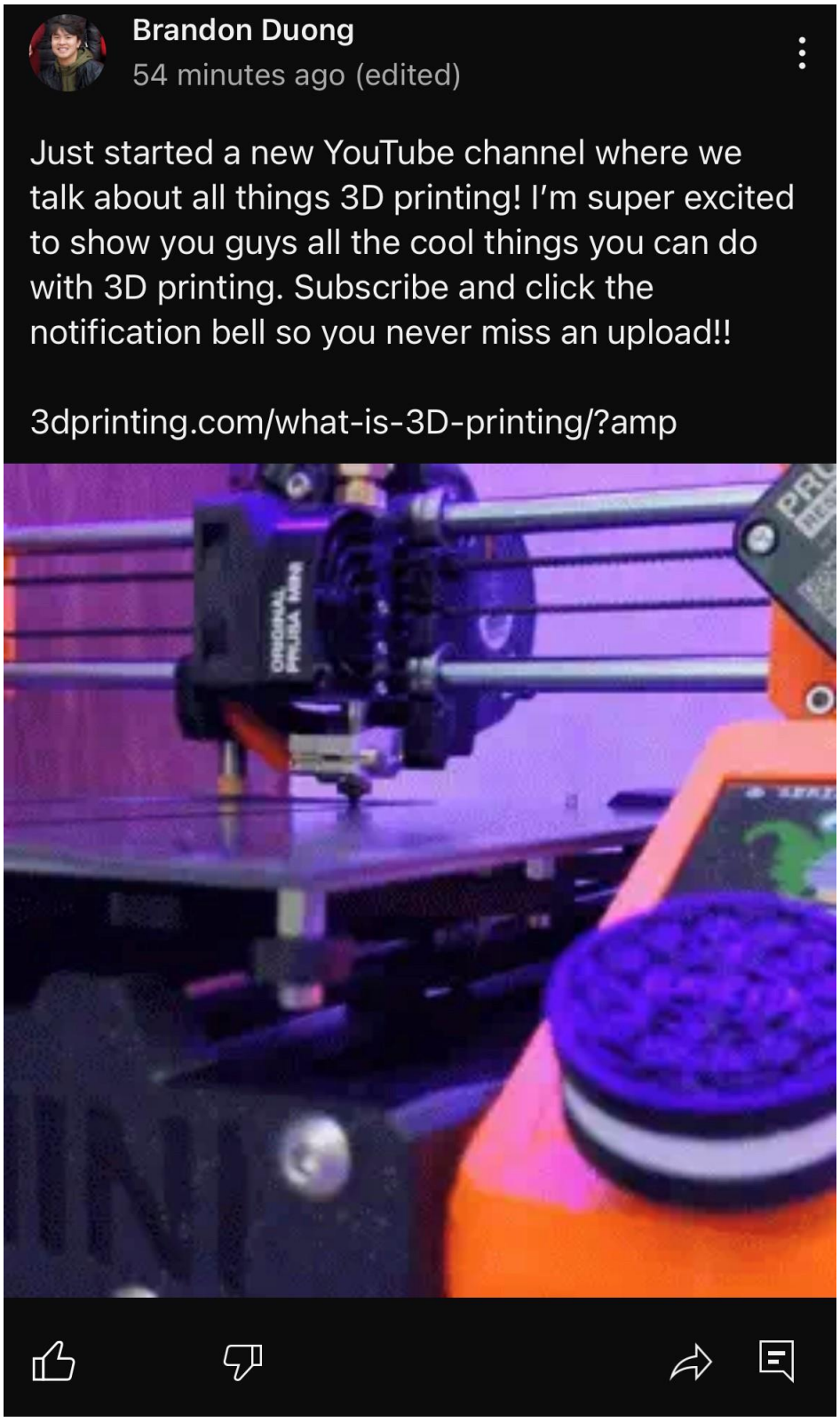
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Brandon Duong

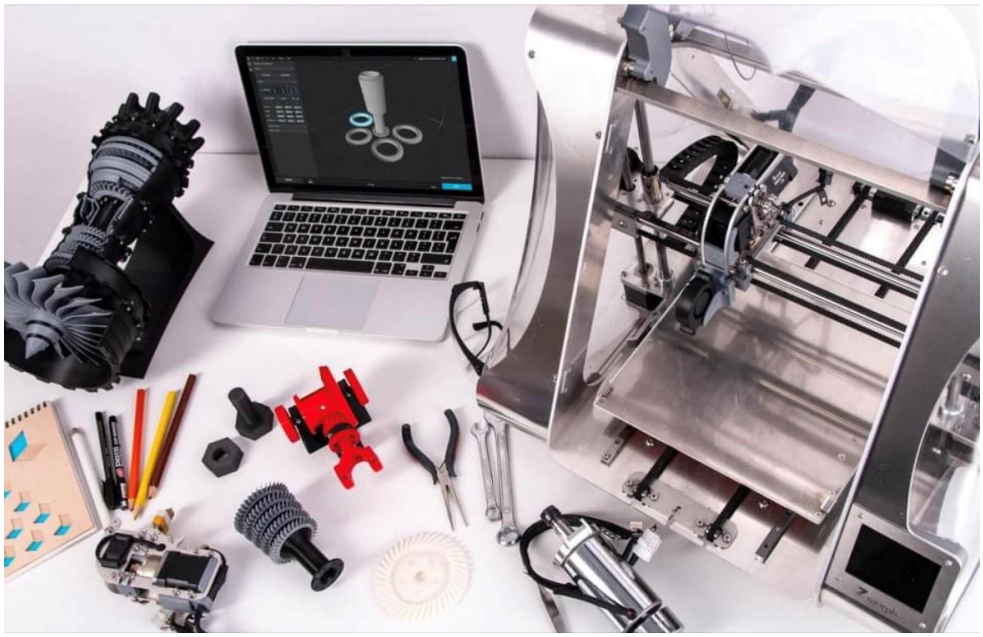
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If you're interested in the 3D printing industry and the future of 3D printing then check out my YouTube in the link! Comment ideas you'd like to see us make videos on!!

https://youtube.com/channel/UCJPyEVHIjZx1WGesixBh5jw

If you're interested on a quick read as well check out the link below from the '3D Printing Industry' about where the industry is going.

https://3dprintingindustry.com/news/the-future-of-3d-printing-additive-manufacturing-experts-forecast-the-next-decade-202831/



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